PG No. 20 $D_{3\mathrm{d}}$ $\bar{3}m$ (-3m1 setting) [trigonal]

Table 1: Representation matrices

Irrep.						
A_{1g}	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{110}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{110}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1\end{bmatrix}$
A_{2g}	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{110}:\begin{bmatrix}-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{110}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-1\end{bmatrix}$
E_g	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$2_{110}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$2_{100}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{110}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
A_{1u}	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{110}:\begin{bmatrix}1\end{bmatrix}$	$2_{100}:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{110}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-1\end{bmatrix}$
A_{2u}	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{110}:\begin{bmatrix}-1\end{bmatrix}$	$2_{100}:\begin{bmatrix}-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{110}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1\end{bmatrix}$
E_u	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$2_{110}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$2_{100}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{110}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1&0\\0&1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$